The Grassroots-to-Mastery Roadmap

Phase 3: Core Harmonics, Neutral Dynamics, and Topological Containment Mechanisms

A Deep-Dive Mathematical and Physical Treatise on Non-Linear Magnetization, Triplen Oscillations, and Stabilizing Tertiary Windings

Module Objective:

To master the non-linear operational domain of three-phase power transformers. You will mathematically derive the decomposition of peaked and flat-topped waveforms, analyze the zero-sequence constraints of triplen harmonics, solve the problem of neutral shift oscillations in ungrounded star connections, contrast the behavior of three-limb vs. five-limb core geometries, and design delta-tertiary containment topologies to eliminate system instabilities.

Pedagogical Sequence Framework

Phase 3 of 6

May 22, 2026

Contents

1 The Physical Geometry of Magnetic Non-Linearity

Cognitive Thought Process & Critical Insight

1. Faraday's Law establishes that if a transformer is excited by a pure sinusoidal voltage, the core flux MUST be a pure sinusoidal wave to induce a matching counter-EMF ($\vec{V} \approx \vec{E} = -d\Phi/dt$).
2. However, ferromagnetic materials exhibit an inherently non-linear $B-H$ curve dominated by magnetic saturation and hysteresis.
3. Therefore, a pure sinusoidal flux wave cannot be sustained by a pure sinusoidal magnetizing current. We must explore the physical and mathematical relationship between these waveforms.

1.1 The Duality of Saturation: Current vs. Flux Distortion

When a transformer core operates near or beyond the knee of its $B - H$ curve, the relationship between magnetic flux density B (and thus total flux Φ) and the magnetic field intensity H (driven by magnetizing current i_m) ceases to be linear. This non-linearity leads to two mutually exclusive operational modes depending on the constraints of the excitation source and winding topologies:

1. **Sinusoidal Flux Mode (Peaked Magnetizing Current):** If the transformer winding is connected to an infinite, low-impedance voltage source, the terminal voltage enforces a perfectly sinusoidal flux waveform ($\Phi(t) = \Phi_m \sin(\omega t)$). Because of core saturation, as the flux reaches its peak, disproportionately larger amounts of magnetizing current (H) are required to sustain it. This distorts the magnetizing current into a sharp, symmetric **peaked waveform**.

2. **Sinusoidal Current Mode (Flat-Topped Flux):** If the transformer winding is driven by a high-impedance source (or constrained by open-circuit winding parameters where harmonic current paths are blocked), a pure sinusoidal current is forced into the winding ($i_m(t) = I_m \sin(\omega t)$). Near the peak of the current wave, core saturation caps the resulting magnetic flux, flattening its peaks. This distorts the flux into a symmetric **flat-topped waveform**.

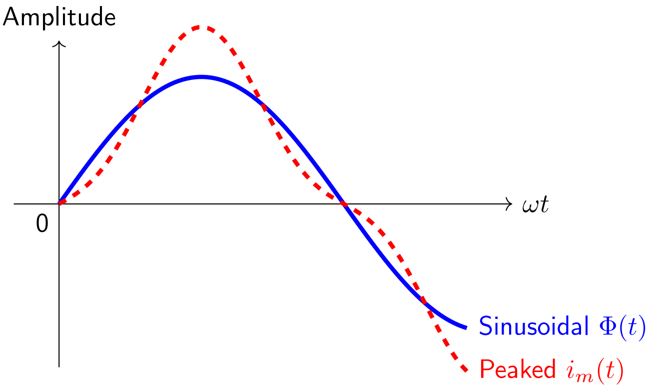

Figure 1: Waveform distortion under sinusoidal flux constraints. The non-linear core saturation demands a peaked magnetizing current rich in odd harmonics.

1.2 Rigorous Mathematical Fourier Decomposition

Because the magnetic material possesses symmetric saturation characteristics for both positive and negative half-cycles, the distorted waveforms exhibit half-wave symmetry:

$$f(t + \pi/\omega) = -f(t) \tag{1}$$

As a direct mathematical consequence, the Fourier expansion of the distorted waveform contains **only odd-order harmonics** (fundamental, 3rd, 5th, 7th, etc.). Even harmonics are identically zero ($a_n = b_n = 0$ for all even n).

Let us write the analytical expression for the peaked magnetizing current required to maintain a sinusoidal flux:

$$i_m(t) = I_{1m}\sin(\omega t) - I_{3m}\sin(3\omega t) + I_{5m}\sin(5\omega t) - I_{7m}\sin(7\omega t) + \dots \tag{2}$$

Conversely, if the magnetizing current is forced to be purely sinusoidal, the flat-topped flux wave decomposes into:

$$\Phi(t) = \Phi_{1m}\sin(\omega t) + \Phi_{3m}\sin(3\omega t) + \Phi_{5m}\sin(5\omega t) + \dots \tag{3}$$

By application of Faraday's Law of Electromagnetic Induction, the induced phase voltage ($e(t)$) resulting from this flat-topped flux is given by:

$$e(t) = -N\frac{d\Phi}{dt} = -N\frac{d}{dt}\left[\Phi_{1m}\sin(\omega t) + \Phi_{3m}\sin(3\omega t) + \Phi_{5m}\sin(5\omega t)\right] \tag{4}$$

$$= -N\omega\Phi_{1m}\cos(\omega t) - 3N\omega\Phi_{3m}\cos(3\omega t) - 5N\omega\Phi_{5m}\cos(5\omega t) \tag{5}$$

Cognitive Thought Process & Critical Insight

`Notice the multiplication factor` n `attached to the amplitude of each harmonic voltage term (`3ω `for the third,` 5ω `for the fifth). Even a minor flat-topped distortion in the core flux wave can induce an extraordinarily high amplitude third-harmonic phase voltage.`

2 The Topological Anomaly of Triplen Harmonics

MANDATORY MULTIMEDIA INTEGRATION: YOUTUBE LECTURE

Video Topic: Transformer Harmonics and Neutral Shift
Search Query / Link: https://www.youtube.com/results?search_query=transformer+harmonics+nptel
Required Focus: Search for visual proofs of why ungrounded systems fail under third harmonic conditions. Watch how the neutral point oscillates.

2.1 Mathematical Proof of Triplen Zero-Sequence Alignment

In a balanced three-phase system, the fundamental components of the magnetizing currents or fluxes are spatially and temporally displaced from one another by exactly 120° electrical. Let us analytically trace what happens to the third harmonic ($n = 3$) components across all three phases (A, B, and C):

$$i_{A3}(t) = I_{3m}\sin(3\omega t) \tag{6}$$

$$i_{B3}(t) = I_{3m}\sin(3(\omega t - 120^\circ)) = I_{3m}\sin(3\omega t - 360^\circ) = I_{3m}\sin(3\omega t) \tag{7}$$

$$i_{C3}(t) = I_{3m}\sin(3(\omega t + 120^\circ)) = I_{3m}\sin(3\omega t + 360^\circ) = I_{3m}\sin(3\omega t) \tag{8}$$

Comparing the equations reveals a unique characteristic:

$$i_{A3}(t) = i_{B3}(t) = i_{C3}(t) \tag{9}$$

The third-harmonic components in all three phases are perfectly synchronized in time and space. They do not form a balanced, phase-displaced three-phase set; instead, they are entirely co-phasor. In symmetrical component theory, this means that **all triplen harmonics ($3^{\text{rd}}, 9^{\text{th}}, 15^{\text{th}}$, etc.) behave strictly as zero-sequence quantities**.

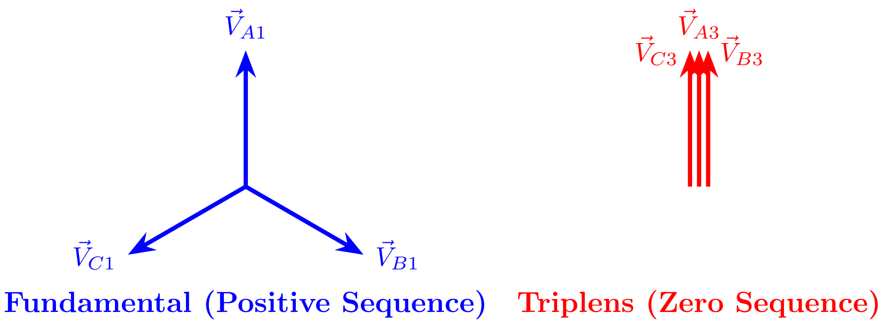

Figure 2: Phasor alignment contrast. Fundamental vectors cancel out symmetrically ($\sum \vec{I}_1 = 0$), whereas triplen vectors add constructively ($\sum \vec{I}_3 = 3\vec{I}_3$).

2.2 Boundary Constraints Governed by Kirchhoff's Laws

Because the triplen harmonic currents are identical in phase across all lines, their physical return paths are heavily bounded by the connection topology of the transformer windings:

- **Star Connection without Neutral Grounding (Y):** Applying Kirchhoff's Current Law (KCL) at the isolated neutral junction N:

$$i_A + i_B + i_C = 0 \tag{10}$$

Substituting the harmonic components, the fundamental and non-triplen currents sum to zero naturally. However, for the triplen components:

$$i_{A3} + i_{B3} + i_{C3} = 3i_{3m} = 0 \implies i_{3m} = 0 \tag{11}$$

Consequently, **triplen harmonic currents can never flow in an ungrounded star connection**.

- **Star Connection with Grounded Neutral (Y_n):** If a neutral conductor exists, KCL at the node yields:

$$i_N = i_A + i_B + i_C = 3I_{3m}\sin(3\omega t) + \dots \tag{12}$$

Triplen currents can now flow, returning collectively through the neutral conductor.

- **Delta Connection (Δ):** In a closed delta loop, the line currents are derived from the differences between phase currents ($i_a - i_b$). Since the triplen currents are identical ($i_{a3} = i_{b3}$), they cancel out completely in the lines:

$$i_{\text{line},3} = i_{a3} - i_{b3} = 0 \tag{13}$$

However, within the closed delta loop itself, the sum of the induced triplen EMFs acts in a continuous loop:

$$\sum \vec{E}_3 = \vec{E}_{ab3} + \vec{E}_{bc3} + \vec{E}_{ca3} = 3\vec{E}_3 \tag{14}$$

This un-canceled voltage drives a continuous **circulating triplen current** around the delta interior, restricting third-harmonic components from leaking into the external transmission lines.

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3 Winding Connections vs. Harmonic Frequency Profiles

3.1 The Isolated Star-Star (Yy) Grid Bottleneck

When both primary and secondary windings are connected in an ungrounded star configuration (Yy), third-harmonic currents are physically blocked from flowing on both sides. This forces the transformer to operate under a **sinusoidal current constraint**.

As derived in Section 1, blocking the peaked components of the magnetizing current forces the core flux to assume a heavily flat-topped waveform. This flat-topped flux wave induces a massive third-harmonic voltage component in each phase winding.

3.2 Deconstruction of the Oscillating Neutral Phenomenon

Because the induced third-harmonic phase voltages ($\vec{E}_{A3}, \vec{E}_{B3}, \vec{E}_{C3}$) are in-phase zero-sequence quantities, they act as a common-mode voltage overlay on the fundamental voltages. Let us map the absolute potential of the physical neutral point n relative to a true reference earth ground:

$$v_n(t) = V_{3m}\sin(3\omega t) \tag{15}$$

This implies that the physical neutral point does not stay at zero volts. Instead, it **oscillates dynamically at three times the grid operating frequency** around the geometric center of the three-phase phasor system.

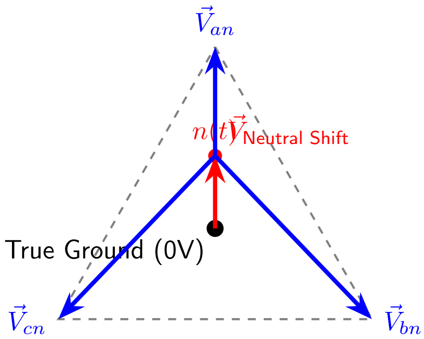

Figure 3: Geometric representation of the Neutral Shift Anomaly. The oscillating neutral point $n(t)$ introduces systematic phase voltage imbalances, creating high insulation stresses.

The consequences of this neutral shift include:

1. **Insulation Stress Profile:** The maximum voltage peak experienced by individual phase winding insulation leaps to $V_{\text{fundamental}} + V_{\text{third}}$. This elevation can cause insulation degradation and catastrophic dielectric breakdown.

2. **Line-to-Line Invariance Immunity:** Interestingly, if we measure the line-to-line voltage ($\vec{V}_{ab} = \vec{V}_{an} - \vec{V}_{bn}$), the zero-sequence components cancel out entirely:

$$\vec{V}_{ab} = (\vec{V}_{A1} + \vec{V}_3) - (\vec{V}_{B1} + \vec{V}_3) = \vec{V}_{A1} - \vec{V}_{B1} \tag{16}$$

   Therefore, the line-to-line voltages remain balanced and sinusoidal, concealing the severe interior phase stress from external measuring equipment connected across the lines.

3.3 The Delta Restoration Mechanism

When at least one winding is connected in Delta (Δ), the closed path allows the induced third-harmonic EMFs to drive a circulating current around the delta loop. This circulating current provides the missing third-harmonic magnetizing MMF needed by the non-linear core.

This self-correcting loop restores the total effective magnetizing current to its required peaked form. As a result, the core flux remains almost perfectly sinusoidal, **neutralizing the oscillating neutral phenomena** and suppressing third-harmonic phase voltages on any accompanying star windings.

4 Geometric Structural Influence of the Iron Core

Cognitive Thought Process & Critical Insight

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1. Up to this point, our mathematical analysis has treated the core flux
as if it exists in a vacuum.
2. In practice, the physical construction of the core structure introduces
structural boundaries that heavily alter zero-sequence flux behavior, even
when using an ungrounded Yy connection.
3. We must evaluate why a 3-phase 3-limb core design behaves differently
from a 5-limb or shell-type structure.
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4.1 The 3-Phase 3-Limb Core Topology

In a standard three-limb core structure, the three magnetic phases occupy three parallel vertical limbs connected across top and bottom common yokes.

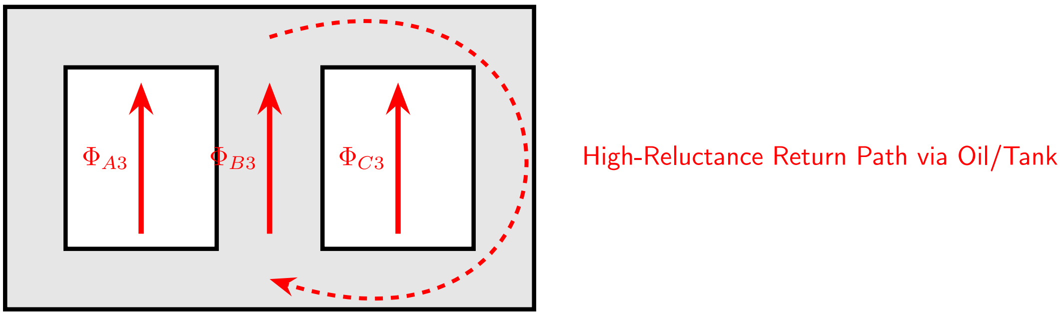

Figure 4: Flux topology of a 3-limb core configuration. Because all three triplen fluxes are in phase, they push upward simultaneously, finding no return path within the iron core.

Because the triplen harmonic fluxes ($\Phi_{A3}, \Phi_{B3}, \Phi_{C3}$) are entirely in phase, they reach their peak upward magnetic orientation at the exact same instant. When they arrive at the top yoke, they cannot clear through neighboring limbs because those limbs are also packed with upward-surging triplen fluxes.

To complete their closed loops, the triplen fluxes are forced to exit the high-permeability iron core structure entirely. They must travel through the high-reluctance paths provided by the surrounding transformer oil, insulation structures, and the structural steel outer enclosure tank wall.

$$\mathcal{R}_{\text{return}} = \mathcal{R}_{\text{oil}} + \mathcal{R}_{\text{tank}} \gg \mathcal{R}_{\text{iron}} \tag{17}$$

Because this non-magnetic loop path has extremely high reluctance, it dampens the magnitude of the third-harmonic flux ($\Phi_3 = \vec{\mathcal{F}}_3/\mathcal{R}_{\text{return}}$) to a negligible fraction (typically less than

2–5% of nominal fundamental flux). Consequently, **in a 3-phase 3-limb core transformer, the neutral shift is naturally suppressed even with an ungrounded Yy winding connection**.

However, the triplen fluxes entering the structural steel tank wall induce eddy currents within it. This causes localized stray heating and lowers the overall efficiency of the machine.

4.2 The 5-Limb and Shell-Type Low-Reluctance Bypass

In a 5-limb core structure, two un-wound outer perimeter fallback limbs are added to provide a return path. In shell-type configurations, the magnetic circuit completely envelopes the windings.

In these structural designs, the upward-surging in-phase triplen fluxes encounter the yokes and find two low-reluctance iron bypass loops ready on either side.

$$\mathcal{R}_{\text{5-limb return}} \approx \mathcal{R}_{\text{iron}} \tag{18}$$

Because the return path is entirely ferromagnetic, the zero-sequence triplen fluxes face very little opposition. They build up to their maximum theoretical saturation limits, distorting the core flux wave into a flat-topped profile. As a result, **a 5-limb or shell-type transformer connected in ungrounded Yy will experience severe neutral shift oscillations and excessive voltage distortion**. These core geometries require a delta winding connection to ensure system stability.

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5 Advanced Design: The Stabilizing Tertiary Winding

Cognitive Thought Process & Critical Insight

1. Suppose an electrical utility company requires a large power transformer to have Star-Star (Yy) connections to establish system grounds on both transmission systems.
2. Furthermore, suppose the core frame design is a 5-limb layout for height restrictions, or a bank of single-phase units.
3. This combination creates high triplen harmonic risks. To fix this without changing the primary or secondary connections, we introduce a third winding: the Delta Tertiary Winding.

5.1 Topological Matrix Integration

A tertiary winding is an auxiliary third winding wound concentrically around the same core limbs alongside the primary and secondary coils. It is connected internally in a closed delta loop (Δ). This winding is isolated from the main input and output power delivery systems.

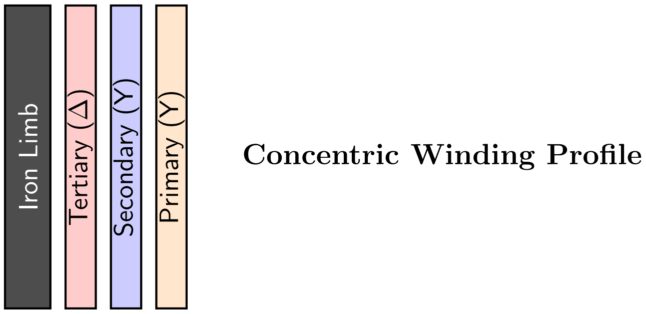

Figure 5: Physical concentric structural stack order on an iron limb core. The delta tertiary layer is placed closest to the core to maximize zero-sequence flux coupling.

The primary (Y_n) and secondary (Y_n) configurations handle fundamental power transfer, while the delta tertiary loop provides a low-impedance path that traps the co-phasor triplen harmonic components.

5.2 Symmetrical Equivalent Zero-Sequence Network Layout

Using per-unit multi-winding notation, the zero-sequence impedance network of a three-winding transformer can be modeled as a multi-branch star equivalent circuit:

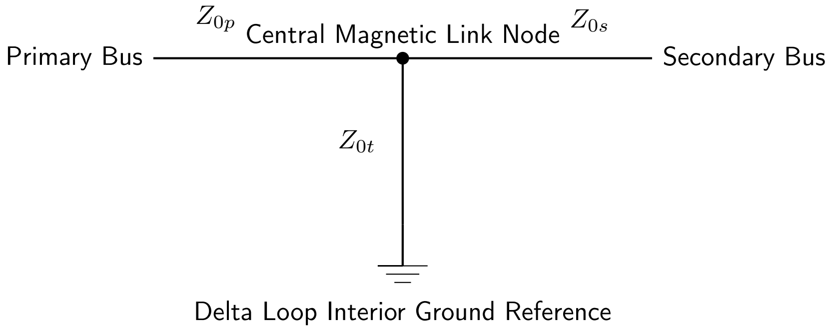

Figure 6: Per-unit zero-sequence equivalent circuit network. The closed delta loop grounds the tertiary branch, shunting triplen harmonic components away from the main systems.

The closed delta connection shorts the tertiary branch to ground within the zero-sequence network. This creates a low-impedance path for triplen frequencies:

$$Z_{0t} \ll Z_{0p}, Z_{0s} \tag{19}$$

The third-harmonic MMF components short-circuit through the tertiary loop, suppressing harmonic voltages across both the primary and secondary output windings.

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6 Rigorous Applied Engineering Problems

6.1 Problem 1: Neutral Displacement Quantification

Scenario Specification: A 3-phase, 5-limb, 11kV/415V, $Yy0$ ungrounded power transformer is energized from an infinite busbar supply. Due to core saturation characteristics under overexcitation conditions, the induced third-harmonic phase EMF inside each winding equals 22% of the fundamental component phase voltage (E_1).

1. Compute the absolute peak voltage stress experienced by the winding insulation layer.
2. Derive the resultant Line-to-Line voltage magnitude and confirm its harmonic content profile.

Step-by-Step Mathematical Solution:

1. Calculate the nominal primary fundamental phase voltage (V_{ph1}):

$$V_{ph1} = \frac{11000}{\sqrt{3}} \approx 6350.85 \text{ V} \tag{20}$$

The third-harmonic voltage component peak value (V_{ph3}) is given by:

$$V_{ph3} = 0.22 \times V_{ph1} = 0.22 \times 6350.85 = 1397.19 \text{ V} \tag{21}$$

Because the fundamental and third-harmonic components have a variable phase relationship that reaches alignment during each cycle, the maximum instantaneous voltage stress across the phase insulation is the direct arithmetic sum of their peaks:

$$V_{\text{stress, max}} = V_{ph1} + V_{ph3} = 6350.85 + 1397.19 = \mathbf{7748.04} \text{ V} \tag{22}$$

This represents a **22% increase in peak dielectric stress** on the insulation assembly compared to a purely sinusoidal baseline.

2. Determine the line-to-line voltage output profile ($\vec{V}_{ab}$):

$$\vec{V}_{ab} = \vec{V}_{an} - \vec{V}_{bn} \tag{23}$$

Expanding this into its fundamental and third-harmonic components:

$$\vec{V}_{ab} = \left(\vec{V}_{A1}\angle 0^\circ + \vec{V}_3\angle 0^\circ\right) - \left(\vec{V}_{B1}\angle - 120^\circ + \vec{V}_3\angle 0^\circ\right) \tag{24}$$

The zero-sequence terms cancel out completely:

$$\vec{V}_{ab} = \vec{V}_{A1}\angle 0^\circ - \vec{V}_{B1}\angle - 120^\circ = \sqrt{3}V_{ph1}\angle 30^\circ \tag{25}$$

$$V_{\text{line-to-line}} = \sqrt{3} \times 6350.85 = \mathbf{11000} \text{ V} \tag{26}$$

The line-to-line voltage remains perfectly sinusoidal and free of third-harmonic distortion, proving that external line-to-line measurements cannot detect the high internal phase stress.

6.2 Problem 2: Circulating Triplen Current Quantification in Delta Loops

Scenario Specification: A 33kV/6.6kV, $Dy1$ core-type transformer has its primary winding connected in delta. The core is subjected to an over-flux condition that induces a third-harmonic EMF component of 450V per phase inside the primary loop. The internal leakage reactance of the winding to the third harmonic frequency ($3\omega = 150$Hz) is evaluated at $12.5\,\Omega$ per phase, and its winding resistance is $0.8\,\Omega$ per phase.

Calculate the magnitude of the circulating current confined within the delta perimeter loop, and find the resulting power loss heating rate.

Step-by-Step Mathematical Solution:

1. Determine the total resultant third-harmonic driving voltage acting around the closed delta loop ($\sum E_3$):

$$\sum E_3 = E_{A3} + E_{B3} + E_{C3} = 3 \times E_3 = 3 \times 450 = 1350 \text{ V} \tag{27}$$

2. Compute the total loop impedance opposing the third harmonic current flow ($Z_{\text{loop, 3}}$):

$$Z_{\text{phase, 3}} = R + jX_3 = 0.8 + j12.5\ \Omega \tag{28}$$

$$Z_{\text{loop, 3}} = 3 \times Z_{\text{phase, 3}} = 3 \times (0.8 + j12.5) = 2.4 + j37.5\ \Omega \tag{29}$$

3. Calculate the absolute magnitude of the circulating loop current ($I_{\text{circulating, 3}}$):

$$I_{\text{circulating, 3}} = \frac{\sum E_3}{|Z_{\text{loop, 3}}|} = \frac{1350}{\sqrt{(2.4)^2 + (37.5)^2}} = \frac{1350}{37.577} \approx \mathbf{35.93} \text{ A} \tag{30}$$

4. Calculate the localized harmonic I^2R power loss heating rate inside the delta winding assembly ($P_{\text{loss, 3}}$):

$$P_{\text{loss, 3}} = (I_{\text{circulating, 3}})^2 \times R_{\text{loop}} = (35.93)^2 \times (3 \times 0.8) = 1290.96 \times 2.4 \approx \mathbf{3098.3} \text{ W} \tag{31}$$

This circulating current generates **3.1 kW of continuous additional heat loss** within the transformer tank, which must be managed by the cooling oil system.

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7 Phase 3 Review and Self-Evaluation

Cognitive Thought Process & Critical Insight

`Verify your engineering competency profile against the technical checklist below before advancing to Phase 4 (Transient Phenomena and Tap Changing Matrices).`

7.1 Theoretical Competency Checklist

1. **Waveform Duality:** Can you explain why a sinusoidal voltage excitation profile forces a peaked magnetizing current wave under core saturation conditions?

2. **Zero-Sequence Matrix Proof:** Show mathematically why third-harmonic components across three symmetrical phases behave as in-phase co-phasors.

3. **Neutral Shift Mechanics:** Describe how an ungrounded neutral point $n(t)$ shifts dynamically in space, and explain why line-to-line measurements do not show this distortion.

4. **Core Geometry Filtering:** Contrast how a 3-limb core frame suppresses triplen fluxes compared to a 5-limb or shell-type design.

5. **Tertiary Stabilization Topology:** Sketch the zero-sequence equivalent network circuit for a three-winding transformer, and explain how a delta tertiary winding traps harmonic components.

You have successfully completed Phase 3. You have mastered non-linear core harmonics, neutral shift dynamics, and topological containment mechanisms. You are now fully prepared to advance to Phase 4: Transient Dynamics, Inrush Currents, and Tap Changing Topology Matrices.